

Data Transmission, Storage, and Information Theory

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- Data Transmission: sending information from one place to another
- Data Storage: storing information for later use
- Information Theory: the study of how pack information into bits



What is Digital Data?

- Not just 0s and 1s (binary)
- Digital means discrete levels
- Analog signals: continuous values
- Digital signals: discrete levels
- Quantization: converting analog to digital
- Sampling rate: how often we measure



Bits per second vs. Baud rate

- Bit rate: bits transmitted per second (bps)
- Baud rate: symbol changes per second
- One symbol can encode multiple bits
- Example: 4-level signal = 2 bits per symbol
- $bps = \text{Baud rate} \times \text{bits per symbol}$



Serial and Parallel Transmission

- Serial: one bit at a time on single wire
- Parallel: multiple bits simultaneously
- Serial: simpler, slower, fewer wires
- Parallel: faster, more wires, synchronization issues
- Modern systems favor serial (PCIe, USB, SATA)



Synchronization

- Clock signal: timing reference
- Synchronous: shared clock between devices
- Asynchronous: no shared clock, start/stop bits
- Clock recovery: extracting timing from data
- Critical for reliable data transfer

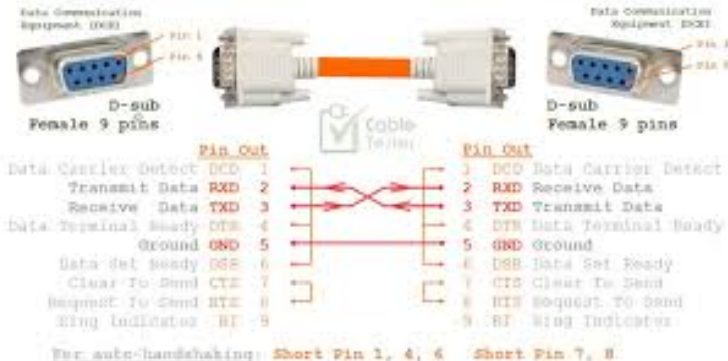


Synchronization Example: RS-232

- 1 start bit
- 8 data bits
- 1 stop bit

The start and stop bits are always there
This helps with synchronization

RS232 Cross Cable (Null Cable) Connection



E

Noise, Cross-talk, and Error Correction

- Noise: random interference
- Cross-talk: signal interference between wires
- Error detection: parity, checksums, CRC
- Error correction: Hamming codes, Reed-Solomon
- Forward Error Correction (FEC): correct without retransmission



- Shared communication pathway
- Multiple devices connected to same wires
- Arbitration: deciding who transmits
- Address/data multiplexing
- Examples: PCI, ISA, memory busses



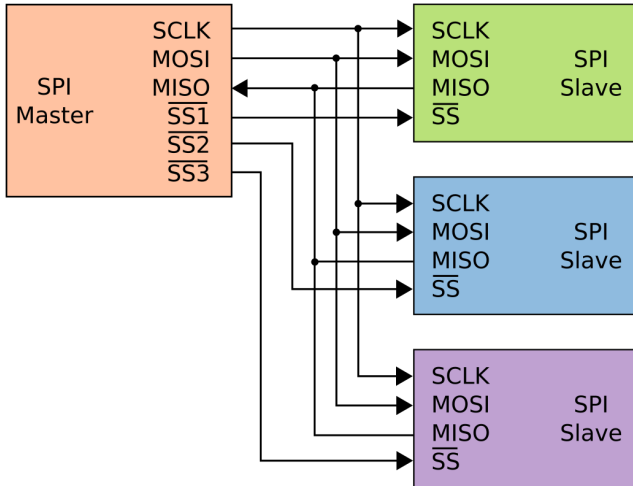
- Serial communication standard
- Asynchronous, point-to-point
- Voltage levels: $\pm 3V$ to $\pm 15V$
- Common for legacy serial ports
- Max distance: ~ 15 meters
- RS-422: 10 Mbps, 1200 meters



- Serial Peripheral Interface
- Master-slave architecture
- 4 wires: MOSI, MISO, SCK, CS
- Synchronous, full-duplex
- Common for sensors, flash memory
- Max distance: 10 meters
- Not standardized, but kHz to MHz speeds are common

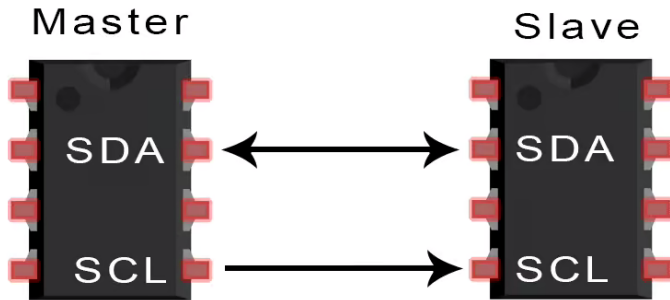


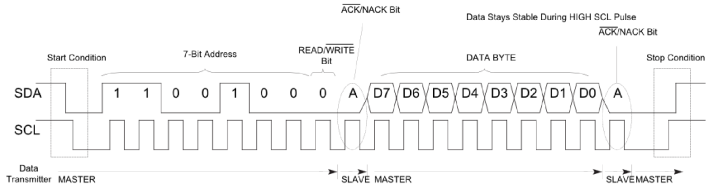
SPI



- Inter-Integrated Circuit
- Two-wire: SDA (data), SCL (clock)
- Multi-master, multi-slave
- Address-based device selection
- Common for low-speed peripherals
- Max distance: 10 meters
- 100 kHz clock frequency
- Fast mode: 400 kHz, Higher speeds up to 3.4 MHz



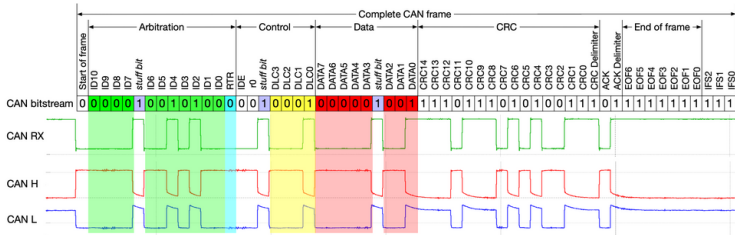




- Controller Area Network
- Automotive and industrial applications
- Multi-master, message-based protocol
- Differential signaling for noise immunity
- Priority-based arbitration



CANbus



- Differential signaling standard
- Balanced transmission lines
- Better noise immunity than RS-232
- Longer distances: up to 1200 meters
- One driver, multiple receivers



- Multi-point differential standard
- Multiple drivers and receivers
- Up to 32 devices on one bus
- Distances up to 1200 meters
- Common for industrial networks

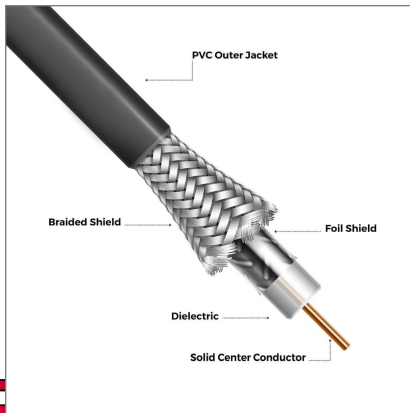


- LAN standard (IEEE 802.3)
- Originally 10 Mbps, now up to 480 Gbps
- Shared wire: each devices has to take turns
- CSMA/CD (Carrier Sense Multiple Access)
- Switched Ethernet eliminates collisions
- TCP/IP protocol stack



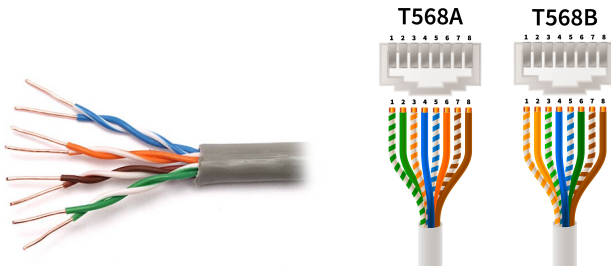
Early Ethernet

- Coaxial cable: 10 Mbps, expensive
- Shielded for noise reduction
- One wire shared by all devices
- Collisions possible, wait a random time and retry



Later Ethernet: Unshielded Twisted Pair

- Unshielded twisted pair: 100 Mbps, cheaper
- Each device has its own wire
- No collisions/No need to wait/retry
- Full-duplex: Both sides can send/receive at the same time.



Later Ethernet: Switches

- Switches: 100 Mbps, 1 Gbps, 10 Gbps
- Each device has its own wire
- No collisions/No need to wait/retry
- Full-duplex: Both sides can send/receive at the same time.
 - Hybrid circuits (to subtract out the node's own transmitted signal)
 - Echo cancellation
 - Near/far-end crosstalk cancellation
 - DSP-based equalization



Modern Ethernet: Switches



- Universal Serial Bus
- Hot-pluggable, power delivery
- USB 1.1: 12 Mbps, USB 2.0: 480 Mbps
- USB 3.0:
- USB 3.1: 10 Gbps
- USB 3.2: 20 Gbps
- USB 4: 40 Gbps
- USB-C: reversible connector



USB CONNECTOR TYPES



- PCI Express
- Serial bus replacing parallel PCI
- Point-to-point links, not shared bus
- Lanes: x1, x4, x8, x16
- PCIe 4.0: 16 GT/s per lane



- Persistent: data survives power loss
- Volatile: data lost without power
- Capacity vs. speed tradeoff
- Access patterns: sequential vs. random
- Storage hierarchy: cache, RAM, disk



Sequential and Random Access

- Sequential: data accessed in order
- Random: any location directly accessible
- Sequential: tape drives, streaming
- Random: RAM, SSDs, hard drives
- Random access faster but more complex



Persistent vs. Volatile Memory

- Volatile: DRAM, SRAM (needs power)
- Persistent: flash, HDD, SSD, ROM
- Non-volatile RAM: battery-backed RAM
- Flash: fast but limited write cycles
- Magnetic: slower but unlimited writes
- All technologies have a capacity vs. speed tradeoff
- All current technologies have a persistence problem (Data erodes)



Static RAM (highest speed)

- delay proportional to $\log_2(size)$
- 6 transistors per bit
- 100 ps delay
- 100 ps delay per bit
- registers: 16 or 32



DRAM (Complicated! and not really Random)

- 2 transistors per bit
- Still very high throughput, tradeoff slower latency
- must be refreshed every few milliseconds
- Arranged in a 2D array of cells
- Latest generation: DDR5
- Row Access ($t_{RAS} = 46$ clocks)
- Column Access ($t_{CAS} = 45$ clocks)
- Precharge ($t_{RP} = 45$ clocks)
- Access a random location: $t_{RAS} + t_{CAS} = 91$ (minimum!)
- burst mode: 16 locations
- Most computers have 2 banks of memory, each 64 bits wide
- 32 sequential accesses: $91 + 31 = 122$ clocks minimum



DRAM DDR5



Block Devices: much slower

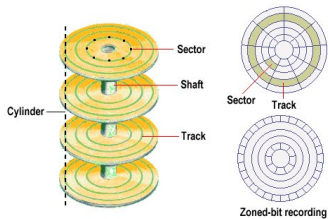
- Latency is much higher, so access is done in much larger blocks
- Block size: 512 bytes or 4 KB or 16 KB or 64 KB
- Hard drives, SSDs are block devices
- OS reads/writes entire blocks
- File systems manage block allocation



- Magnetic storage, rotating platters
- Read/write heads on actuator arms
- Seek time: moving heads to track (mS)
- Rotational latency: waiting for sector
- Capacity: up to 26+ TB
- Speed can be Gb/s with multiple platters



Hard Drives



- Solid State Drives
- NAND flash memory, no moving parts
- Much faster than HDDs
- Limited write cycles (wear leveling)
- SATA and NVMe interfaces



SSDs



- NAND: storage (SSDs, USB drives)
- NOR: code storage (faster reads)
- Erase-before-write limitation
- Wear leveling extends lifespan
- MLC, TLC, QLC: multiple bits per cell
- Periodic recharge is needed (data erodes over years)
- This limits ultimate lifetime because data is re-written



Tape Drives

- Sequential access storage
- High capacity, low cost per GB
- Slow access, archival use
- LTO (Linear Tape-Open) standard
- Backup and long-term storage



Tape Drives



Optical Drives

- CD (Compact Disc)
- DVD (4.7 GB)
- Blu-ray (25 GB or 50 GB)
- Laser reads/writes pits and lands
- Read-only, write-once, rewritable
- Declining use, replaced by USB/cloud
- Longer lifetime than hard drives, but still limited



- Encode data in DNA sequences
- Extremely high density
- Long-term stability (thousands of years)
- Very slow read/write
- Research stage, not practical yet

Microsoft DNA Storage



Holographic Storage

- Store data in 3D holograms
- High capacity potential
- Parallel read/write operations
- Research and limited commercial use
- Challenges with media stability

